

PRAYAS

2.0 2027

Physics

Lecture No - 02

MOTION IN A STRAIGHT LINE

Manish Singh Tak (Masti Sir)



Topics to be covered

1 Speed and Velocity

2 Sign Convention

3

4



RECAP AND HOMEWORK

The text is surrounded by various colorful educational icons. Above 'RECAP' is a yellow alarm clock. To the left is a notepad with green checkmarks. To the right is a blue calculator and a gold medal with 'A+'. Below 'AND' is a green stapler. Below 'HOMEWORK' are a red pencil, a blue paperclip, a red apple, a wooden ruler, a glue stick, a scroll labeled 'H.W.', and a stack of three books (blue, yellow, and red).

Question

discuss tomorrow



Let $|\vec{A}| = 3$, $|\vec{B}| = 4$ and both vector are at an angle 60° Find value of

1. $\vec{A} \cdot \vec{A}$

2. $\vec{A} \cdot (3\vec{A} + 2\vec{B})$

3. $(\vec{A} + \vec{B})^2$

Question

discuss tomorrow



Let $|\vec{A}| = 6$, $|\vec{B}| = 5$ and $|\vec{A} + \vec{B}| = 9$. The value of $(3\vec{A} + 4\vec{B}) \cdot (\vec{A} - \vec{B})$ is:

Question

discuss tomorrow



if vector $(\vec{A} + 2\vec{B})$ and $(3\vec{A} - \vec{B})$ are perpendicular then find angle between $\vec{A}$ and $\vec{B}$
(Given $A = B$)

(A) $\cos\theta = \frac{2}{5}$

(B) $\cos\theta = \frac{1}{5}$

(C) $\cos\theta = \frac{1}{2}$

(D) $\cos\theta = \frac{1}{4}$

Ans B

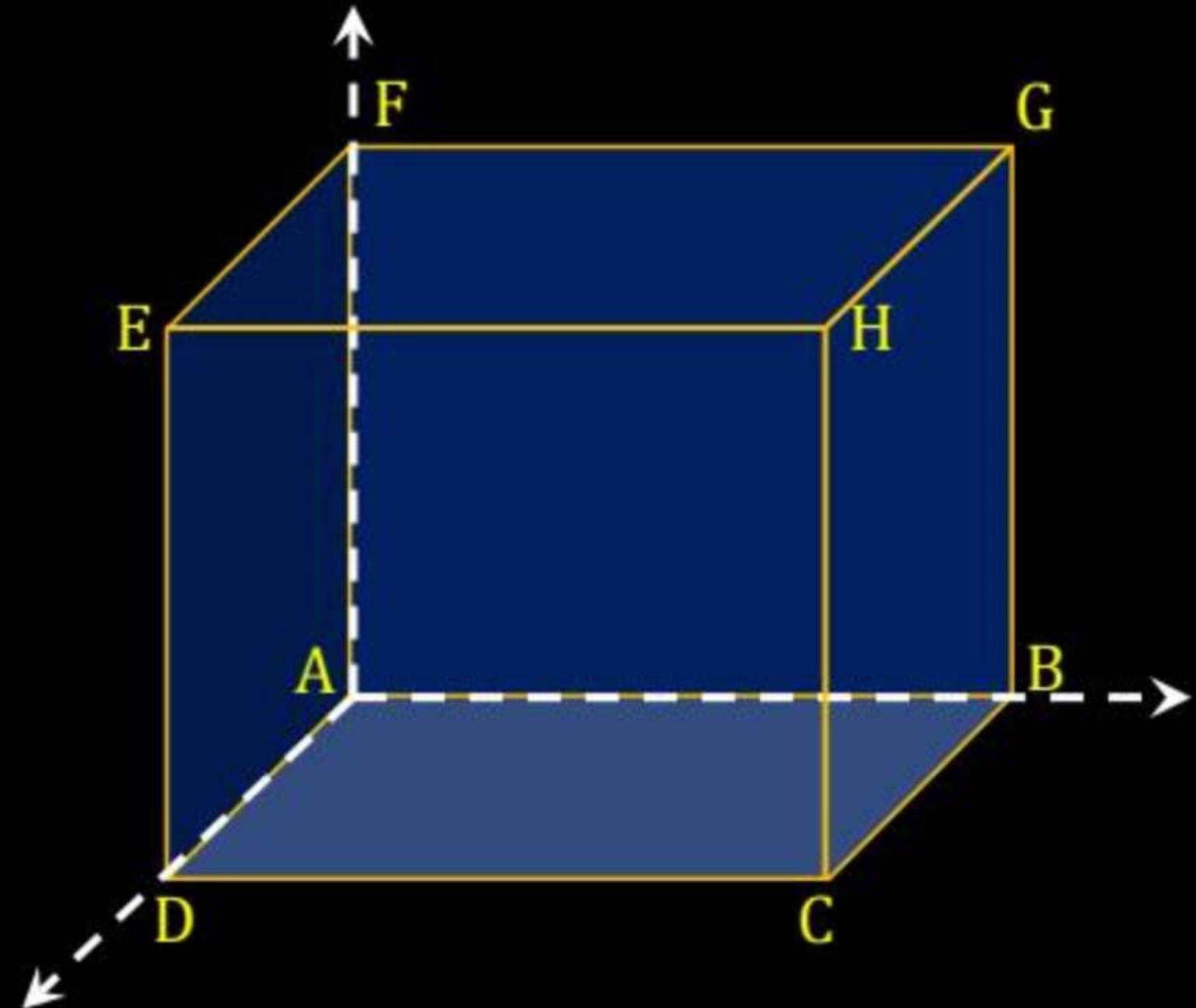
Question

discuss tomorrow



Consider a cube of side a find angle between following vectors

1. $\vec{AH}$ and $\vec{DG}$
2. $\vec{CE}$ and $\vec{CF}$



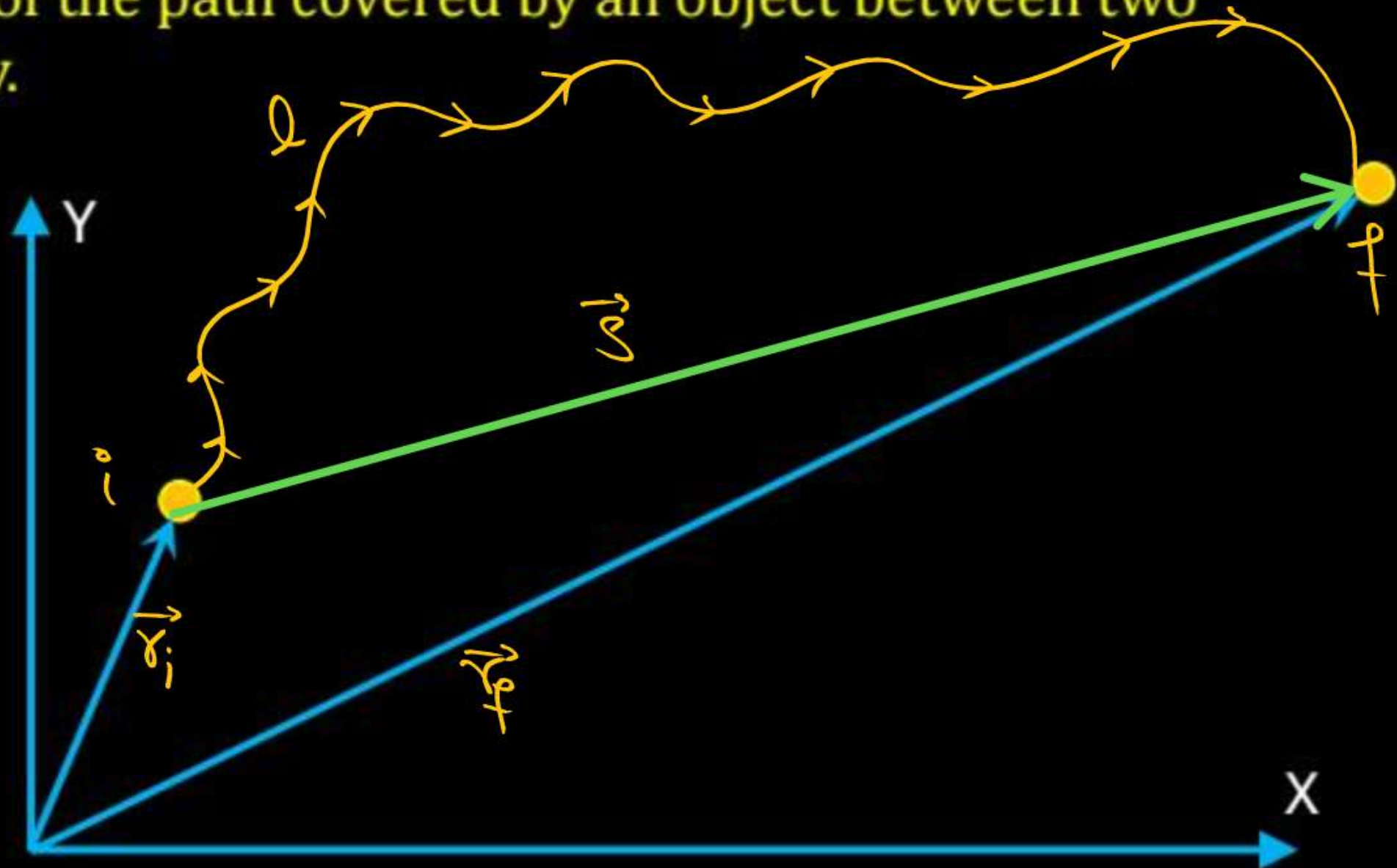
Displacement ($\vec{S}$) and Distance travelled (l)



Displacement ($\vec{S}$):- a vector quantity representing the change in position of an object, defined as the straight-line distance and direction from its initial to final point

Distance travelled (l):- The total length of the path covered by an object between two positions, representing a scalar quantity.

$$\vec{S} = \vec{r}_f - \vec{r}_i$$



Important points



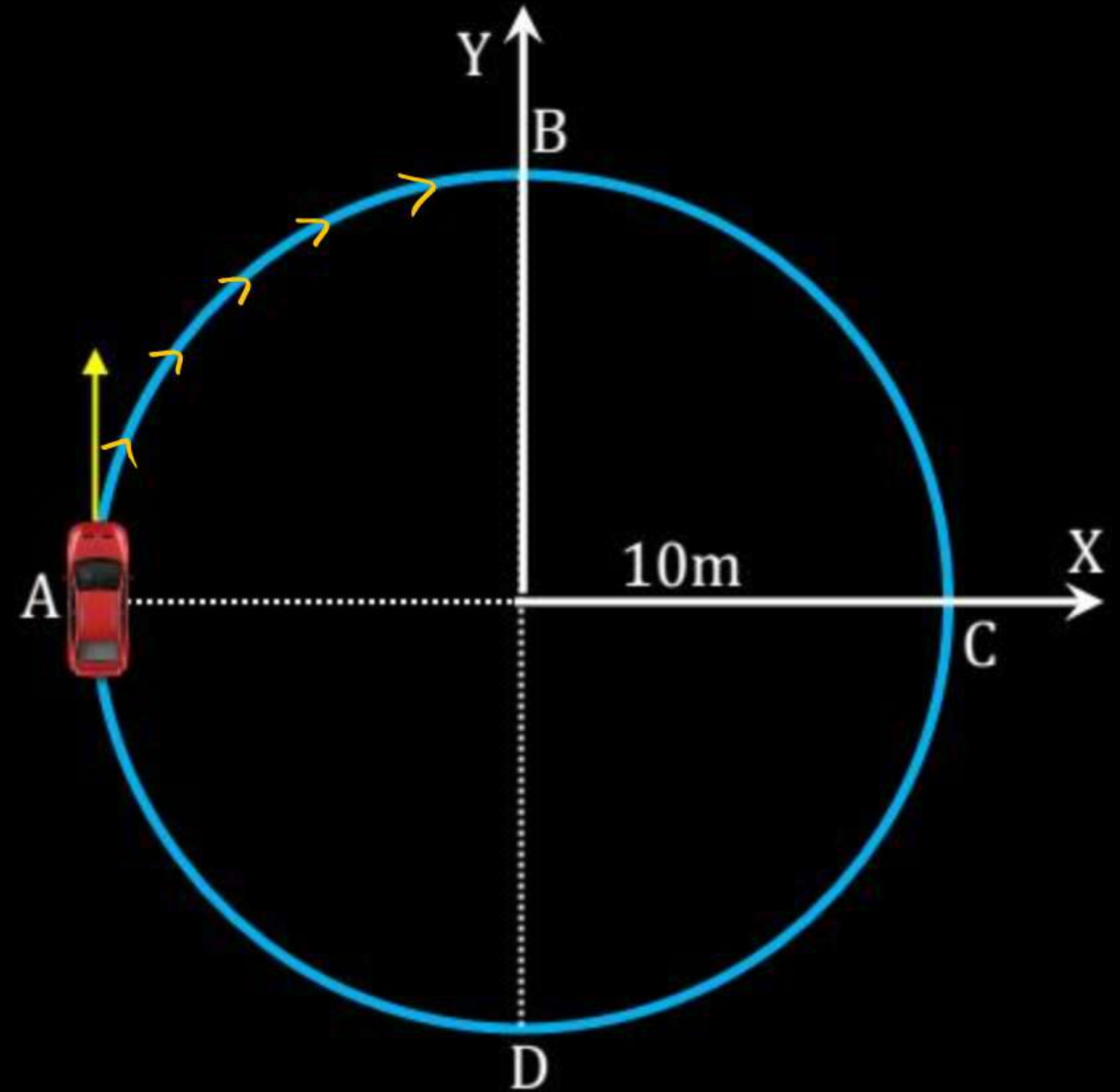
- Displacement($\vec{S}$) is a vector quantity
- $\vec{S} = \vec{r}_f - \vec{r}_i$
- Distance travelled l is a scalar quantity
- Displacement is always less than equal to distance travelled
- Displacement and distance travel does not depend on choice of origin
- Distance travelled can't be negative and can't decreases with time

Question



A car is moving on a circle of radius 10m. Find Displacement $\vec{S}$ and distance travelled l

1. A to B
2. A to C
3. A to D

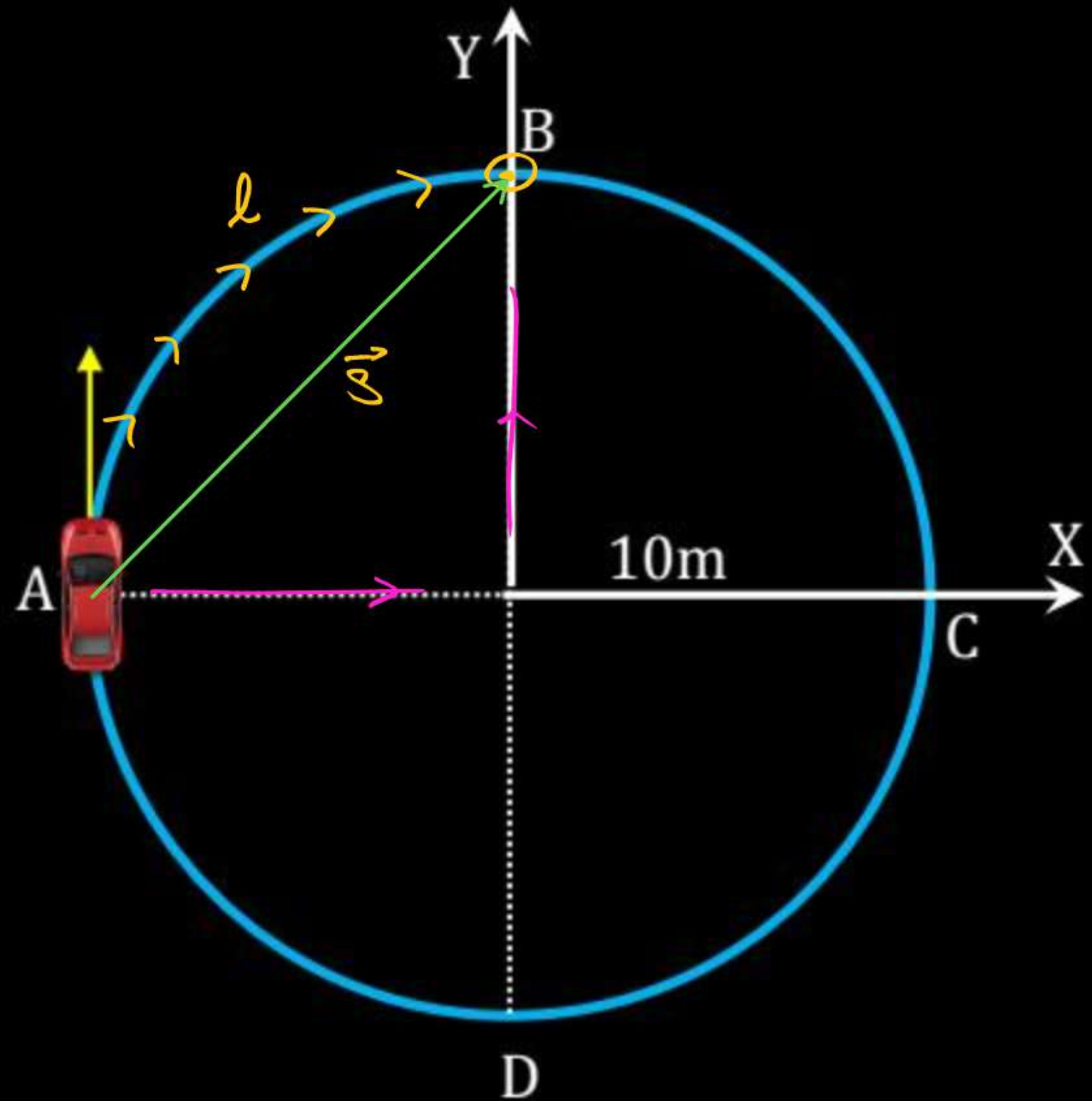


1. A to B

$$l = R\theta = R(\pi/2)$$

$$\vec{s} = R\hat{i} + R\hat{j}$$

$$S = \sqrt{R^2 + R^2} = \sqrt{2}R$$

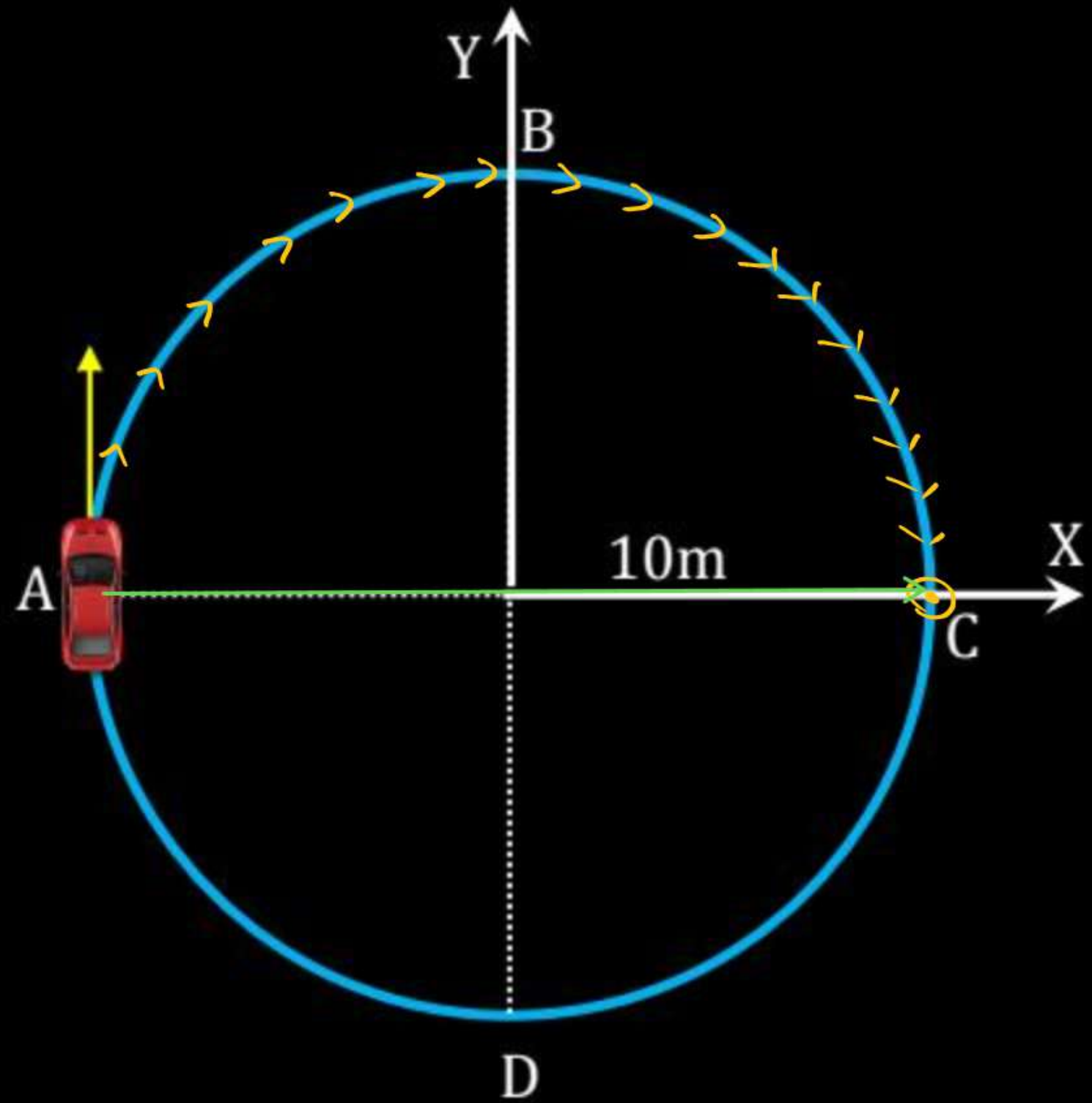


1. A to C

$$l = R\theta = 10(\pi) \text{ m}$$

$$\vec{s} = 2R\hat{i} = 20\hat{i}$$

$$s = 20$$

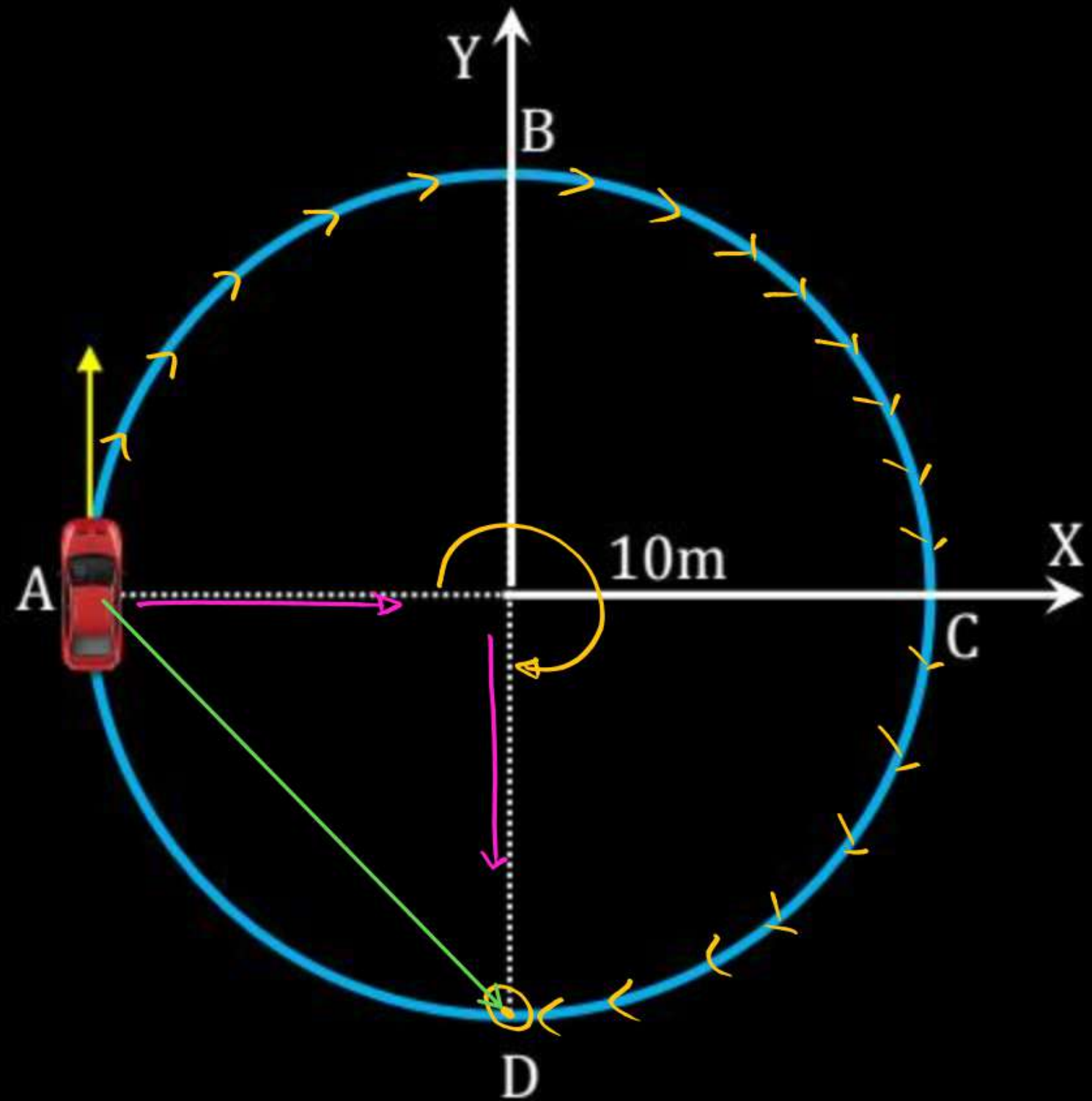


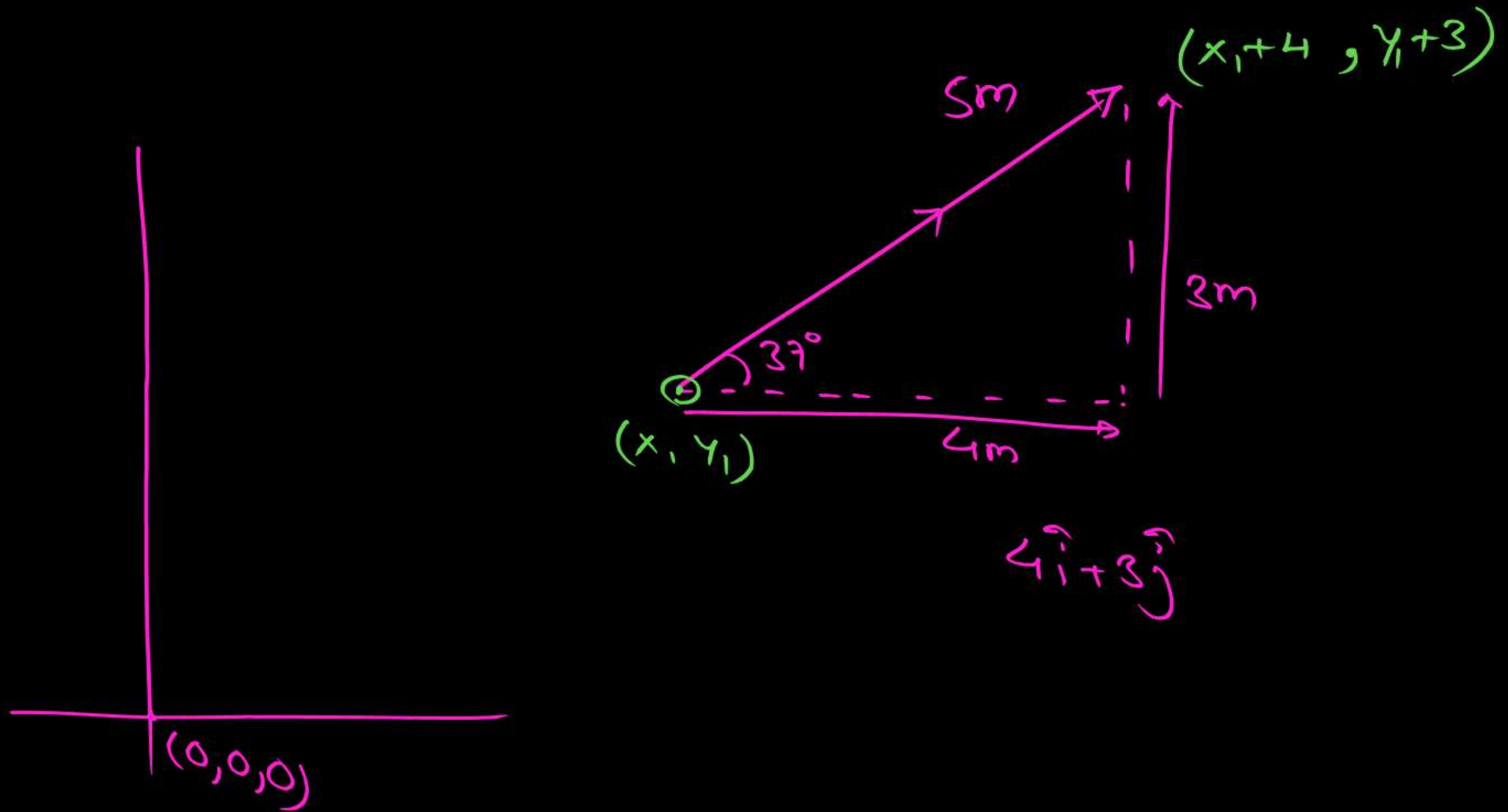
1. A to D

$$l = R\theta = 10\left(\frac{3\pi}{2}\right)$$

$$\vec{S} = R\hat{i} - R\hat{j} = 10(\hat{i} - \hat{j})$$

$$S = \sqrt{10^2 + 10^2} = 10\sqrt{2} \text{ m}$$







Speed (v) and Velocity ($\vec{v}$)

Instantaneous Speed (v) :- Rate of change of distance travelled

$$v = \frac{dr}{dt} = \frac{dl}{dt}$$

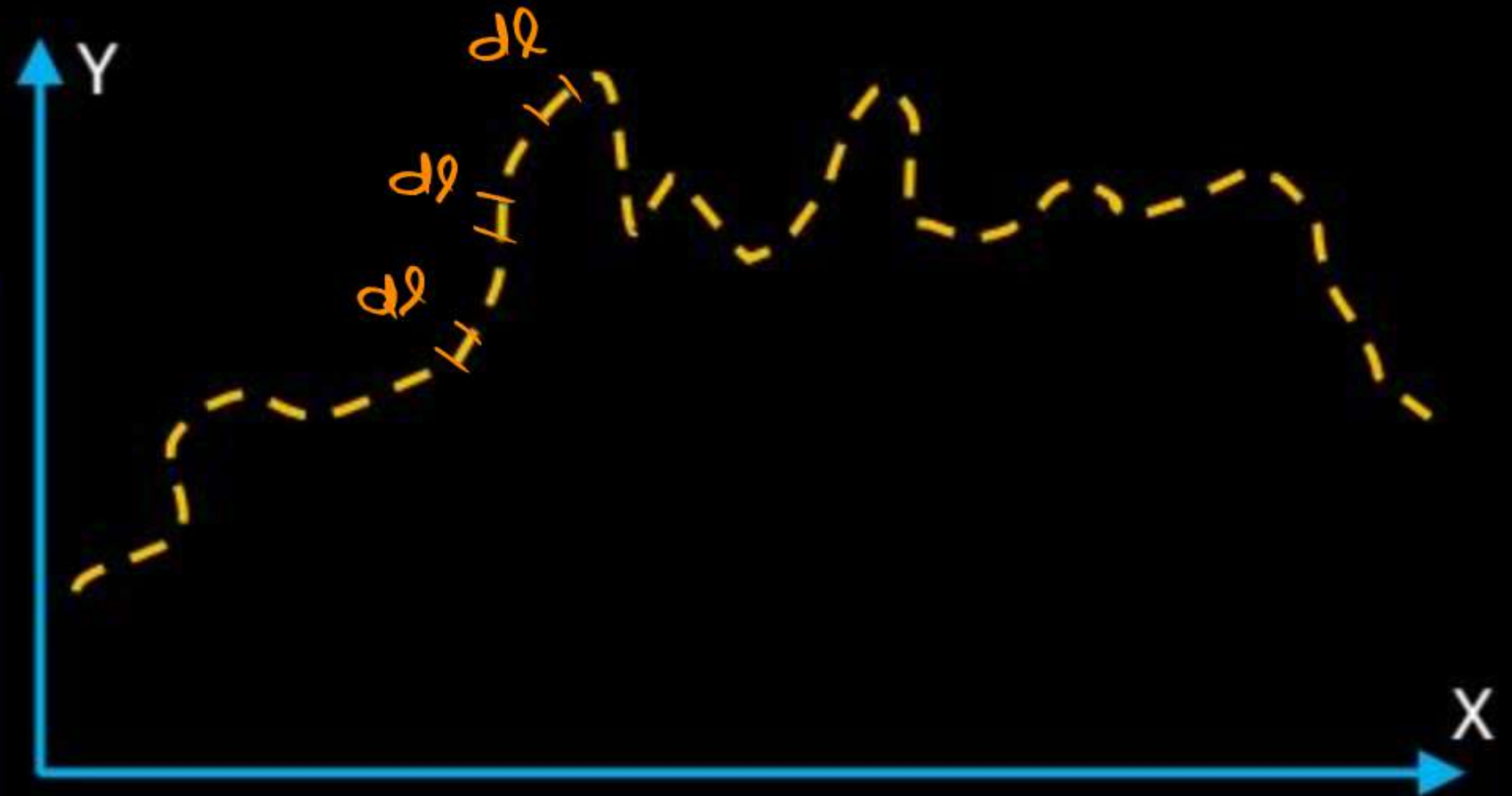
SI unit $\rightarrow$ m/s

(for very small part of path
|displacement| = distance $\Rightarrow dr = dl$
travelled)



$$\left(\begin{array}{c} 36 \text{ kmph} \rightarrow 10 \text{ m/s} \\ \downarrow \\ \frac{36 \times 1000}{3600} \end{array} \right)$$

$$\begin{aligned} \text{Average Speed} &= \frac{\Delta l}{\Delta t} \quad \text{or} \quad \frac{l}{\Delta t} \\ &= \frac{\text{Total distance travelled}}{\text{Total time}} \end{aligned}$$



Instantaneous Velocity ($\vec{v}$) :- Rate of change of position

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d\vec{S}}{dt} = v\hat{v}$$

Speed $\leftarrow$ v dirⁿ $\leftarrow$ $\hat{v}$

$(dr = dl \Rightarrow |velocity| = Speed)$

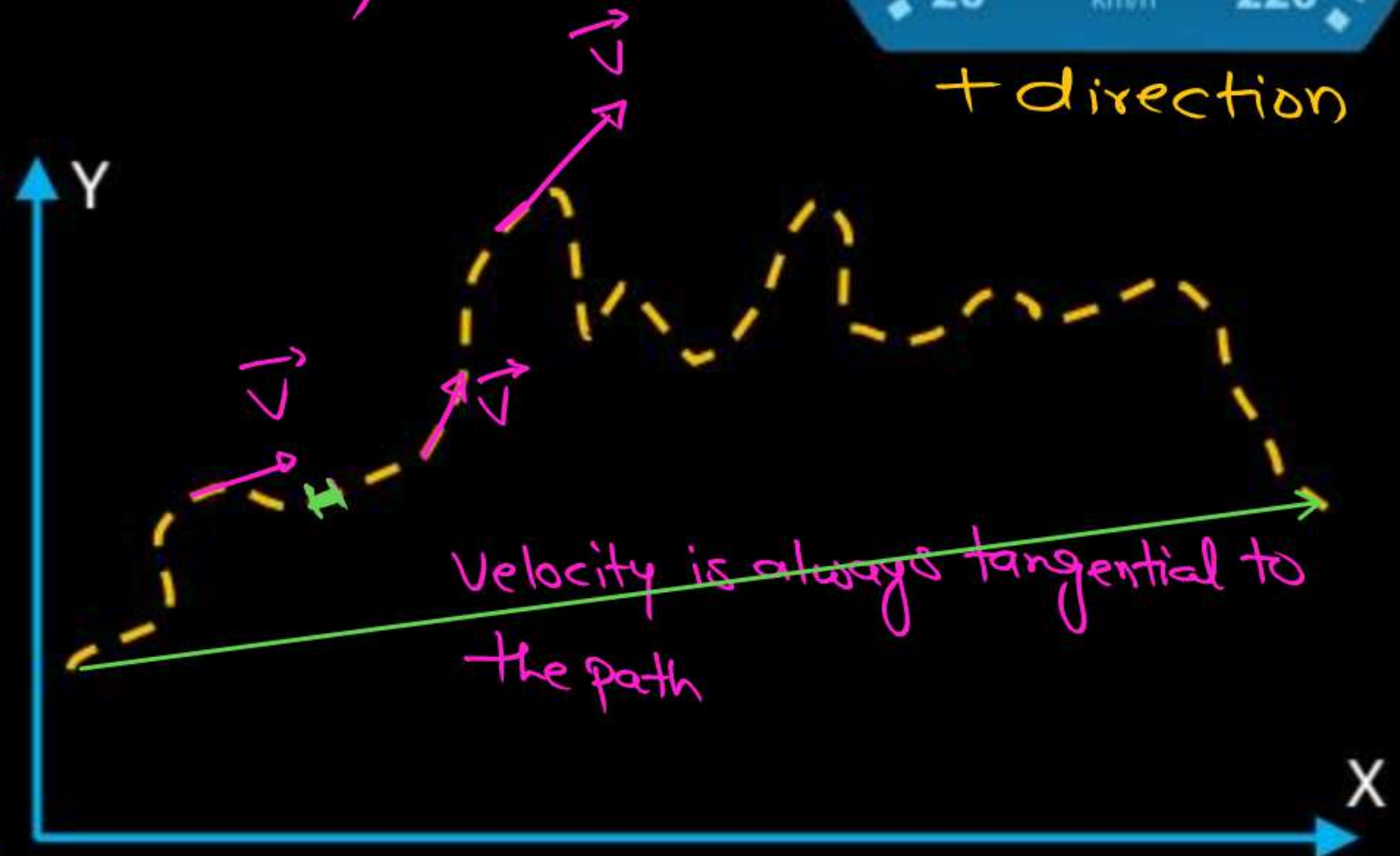


+ direction

Average velocity = $\frac{\Delta \vec{r}}{\Delta t} = \frac{\vec{S}}{\Delta t}$

= $\frac{\text{displacement}}{\text{time}}$

Avg. velocity $\leq$ Avg. Speed



Average Speed and Average Velocity

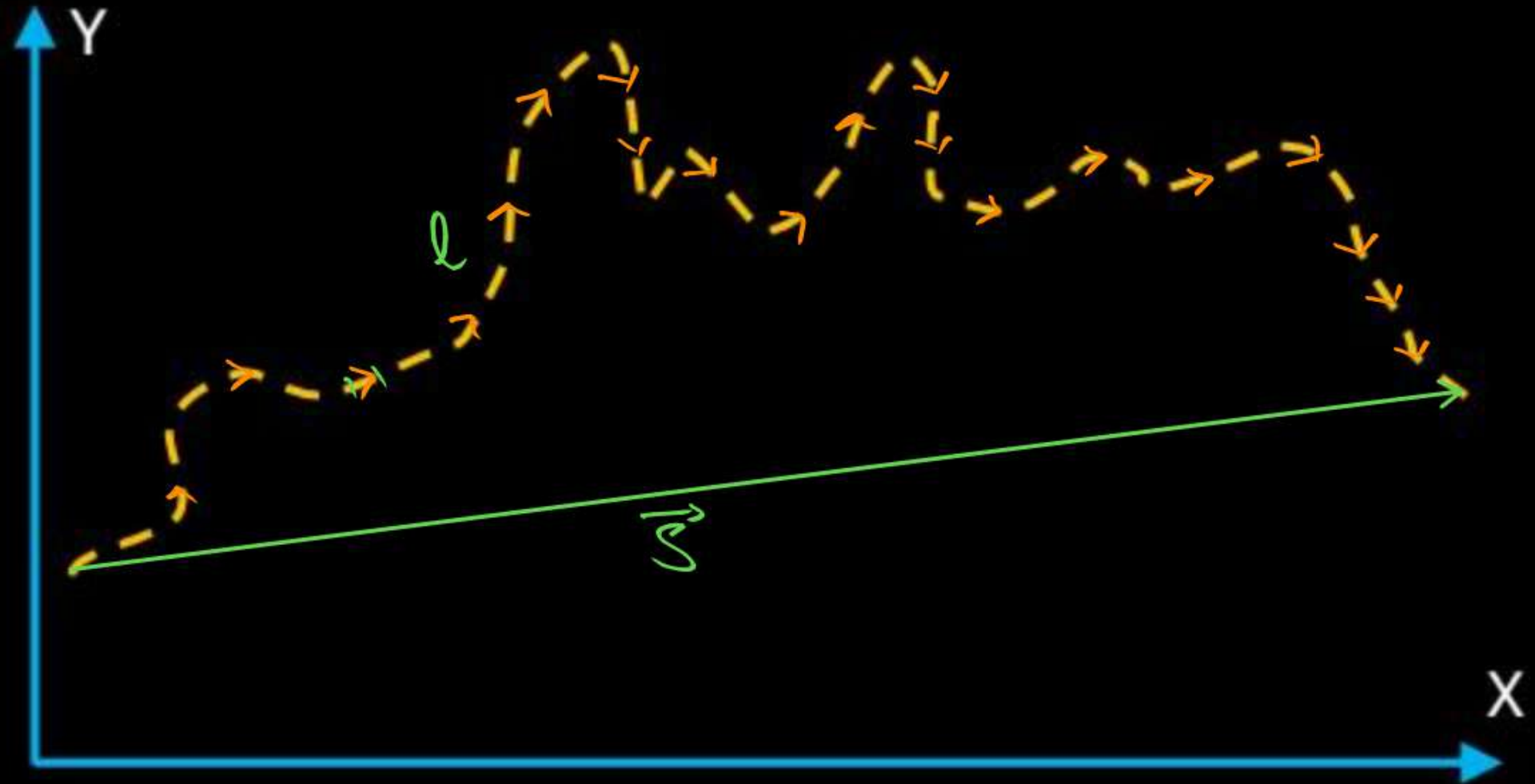


$$\text{Avg. Speed} = \frac{l}{t}$$

$$\text{Avg. velocity} = \frac{\vec{S}}{t}$$

$$S \leq l$$

$$\text{Avg. velocity} \leq \text{Avg Speed}$$



Important points



- Magnitude of Instantaneous Velocity and Instantaneous Speed are same
- Magnitude of Average velocity is always less than equal to average speed

Uniform Velocity Motion

Uniform $\rightarrow$ एक जैसा, एक समान

Uniform velocity $\Rightarrow \vec{V} = \text{Constant}$

magnitude + dirⁿ $\rightarrow$ Constant

Ex. 1

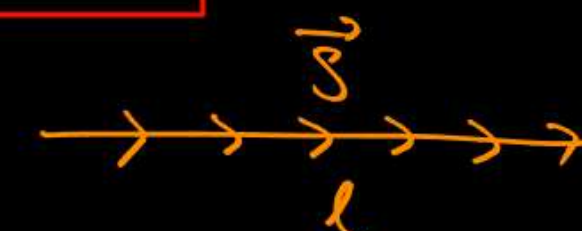


$$20 \text{ m/s} = \text{Constant}$$

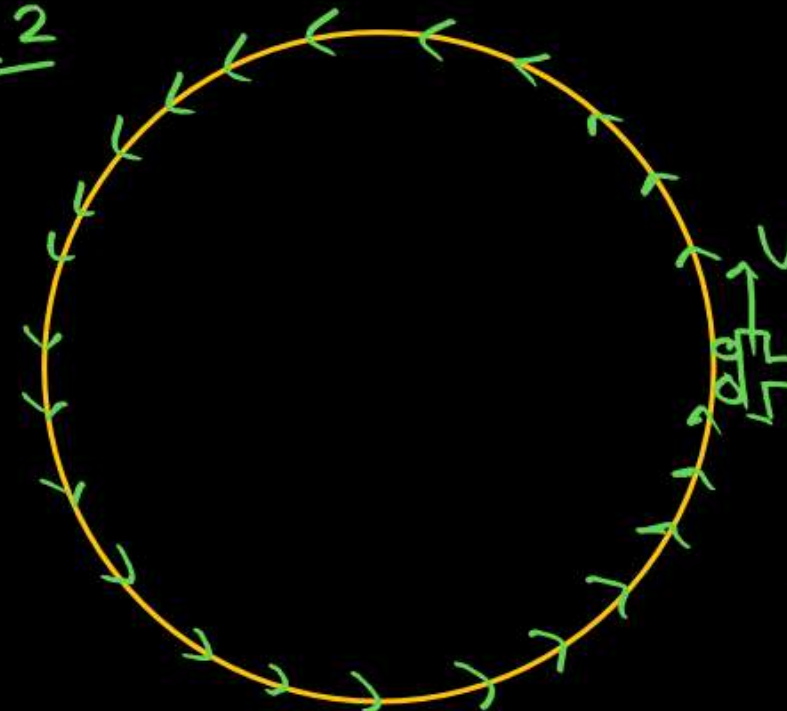
$$\vec{V} = \text{Const.}$$

$$\vec{S} = \vec{V}t = \vec{r}_f - \vec{r}_i$$

$|\vec{S}| = l$



Ex. 2



$$V = 20 \text{ m/s}$$

$$V = \text{Const.}$$

$$\vec{V} \neq \text{Const.}$$

due to dirⁿ

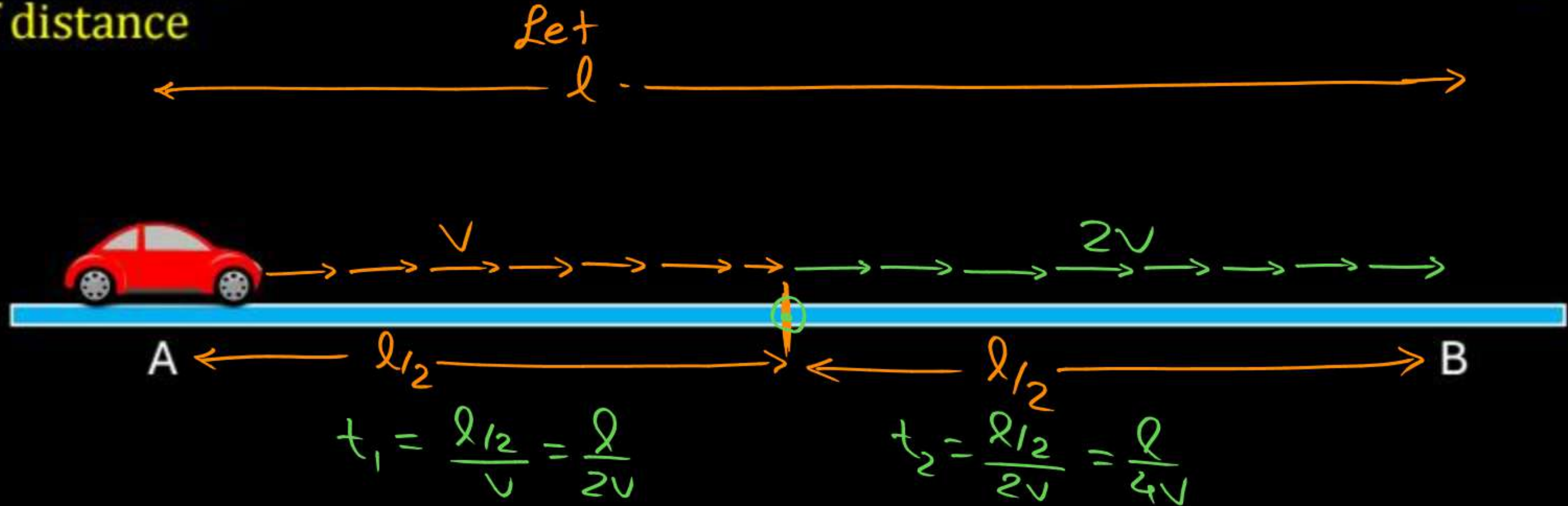
Question



$$S = vt$$



A car is moving on a straight road from point A to point B find average speed of car if Car move with constant speed V for 1st half distance and then car move with constant speed $2V$ for 2nd half distance



$$\text{Avg. Speed} = \frac{l}{t} = \frac{l}{t_1 + t_2} = \frac{l}{\frac{l}{2V} + \frac{l}{4V}} = \frac{l}{\frac{2l + l}{4V}} = \frac{4V}{3}$$

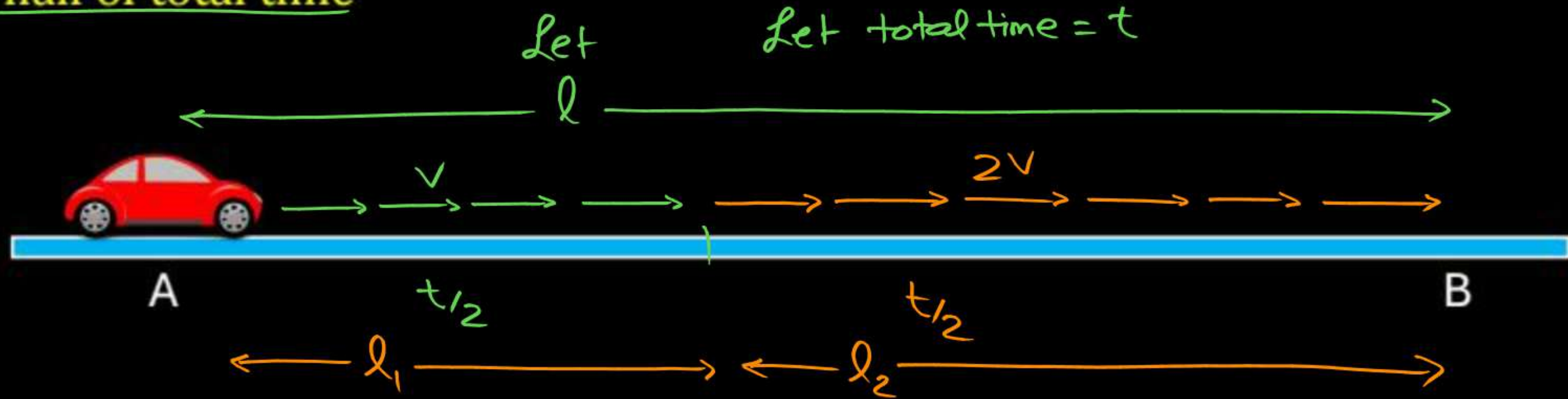
Question

$$S = vt$$

$$V = 60 \text{ m/s} \longrightarrow$$



A car is moving on a straight road from point A to point B find average speed of car if Car move with constant speed V for 1st half of total time and then car move with constant speed $2V$ for 2nd half of total time



$$\text{Avg. Speed} = \frac{l}{t} = \frac{l_1 + l_2}{t} = \frac{V(t/2) + 2V(t/2)}{t} = \frac{3V}{2}$$

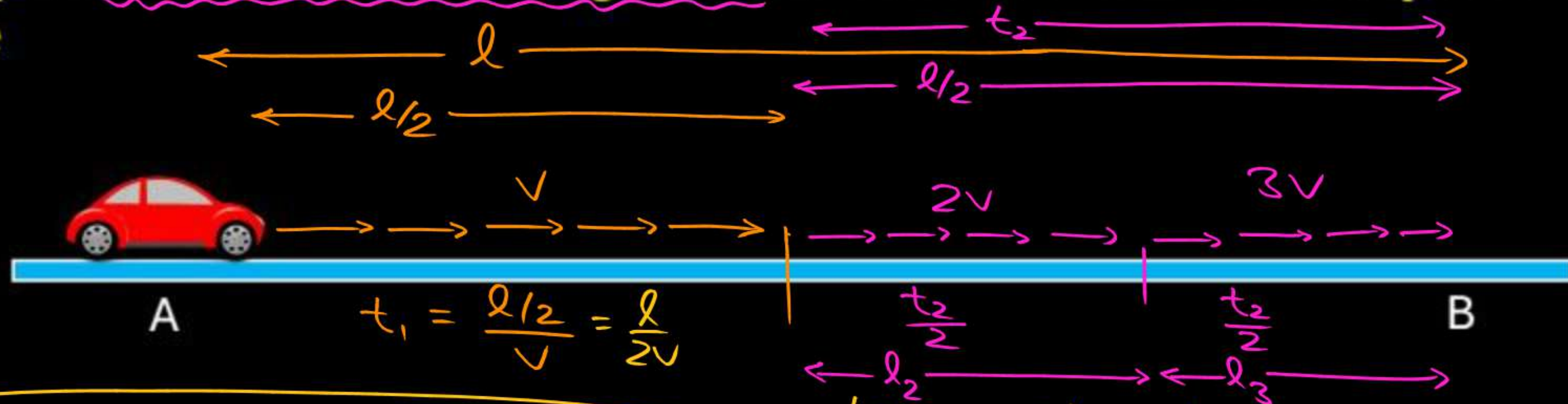
Question

$$S = vt$$

$$V = 42 \text{ m/s}$$



A car is moving on a straight road from point A to point B find average speed of car if Car move with constant speed V for 1st half distance and then for 2nd half distance, car move with constant speed $2V$ for half of remaining time and then move with constant speed $3V$ for rest of the time



$$\begin{aligned} \text{Average Speed} &= \frac{l}{t} = \frac{l}{t_1 + t_2} = \frac{l}{\frac{l}{2V} + \frac{l}{5V}} \\ &= \frac{l}{\frac{5l + 2l}{10V}} = \frac{10V}{7} \end{aligned}$$

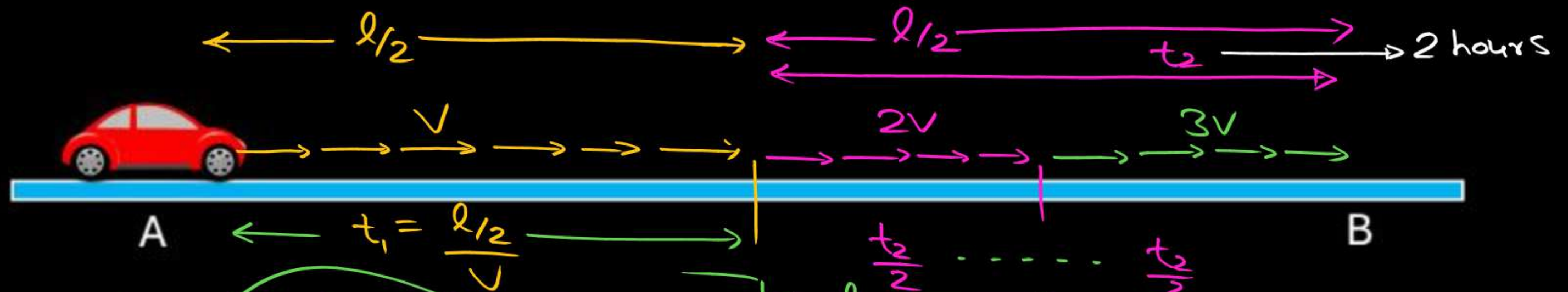
$$\begin{aligned} (l_2 = 2V \cdot \frac{t_2}{2}) \quad (l_3 = 3V \cdot \frac{t_2}{2}) \\ \text{but } l_2 + l_3 = l/2 \Rightarrow 2V \cdot \frac{t_2}{2} + 3V \cdot \frac{t_2}{2} = \frac{l}{2} \\ t_2 = \frac{l}{5V} \end{aligned}$$

Question



$$S = vt$$

A car is moving on a straight road from point A to point B find average speed of car if Car move with constant speed V for 1st half distance and then for 2nd half distance, car move with constant speed $2V$ for half of remaining time and then move with constant speed $3V$ for rest of the time



$$V_{avg} = \frac{l}{t_1 + t_2}$$

$$t_1 = \frac{l/2}{V}$$

$$\frac{l}{\frac{l}{2V} + \frac{l_2}{V}}$$

$$\frac{l}{\frac{l}{2V} + \frac{l_3}{3V}}$$

$$l_2 = 2V \cdot \frac{t_2}{2}$$

$$l_3 = 3V \cdot \frac{t_2}{2}$$

$$l_2 + l_3 = l/2 \Rightarrow 2V \cdot \frac{t_2}{2} + 3V \cdot \frac{t_2}{2} = \frac{l}{2} \Rightarrow t_2 = \frac{l}{5V}$$

Acceleration ($\vec{a}$)

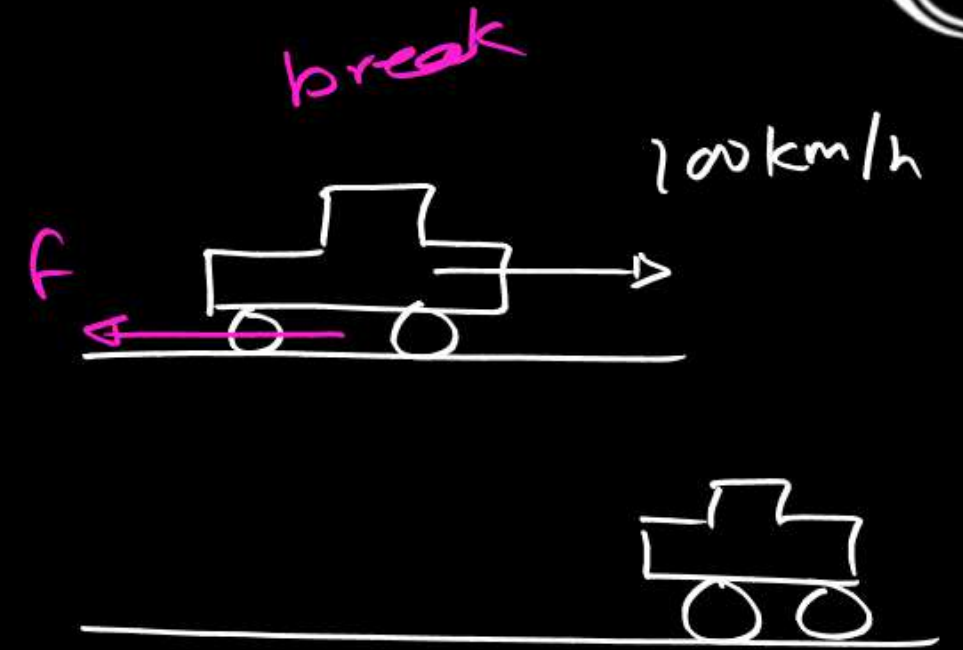
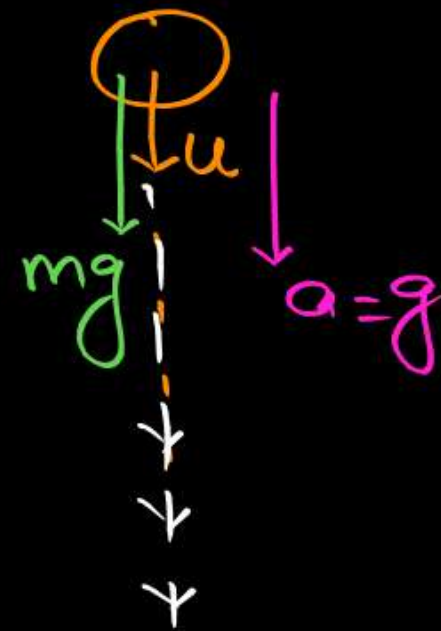
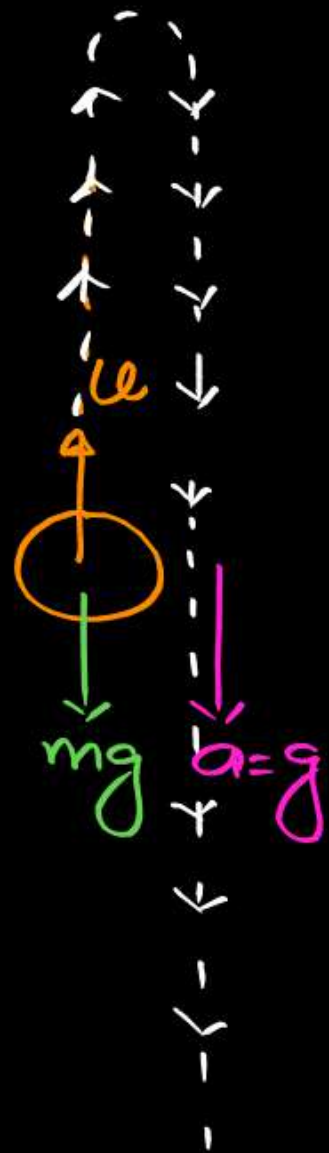
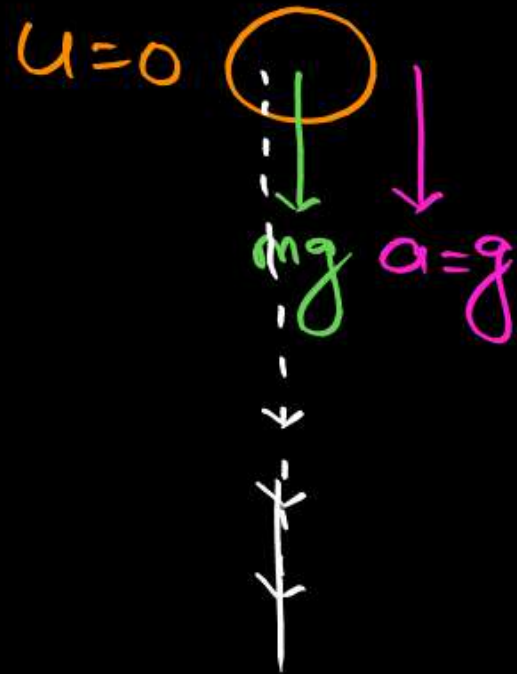
Instantaneous Acceleration ($\vec{a}$) $\rightarrow$ rate of change of velocity

$$\vec{a} = \frac{d\vec{v}}{dt} \quad \text{SI unit } m/s^2$$

- $\rightarrow$ dirⁿ of motion of particle at any instant is only given by velocity at that moment (not by acceleration)
- $\rightarrow$ acc. changes velocity with time
- $\rightarrow$ acc. is always come from force $(\vec{F} = m\vec{a} \Rightarrow \vec{a} = \frac{\vec{F}}{m})$ (dirⁿ of $\vec{F}$ and $\vec{a}$ are same)

Average Acceleration ($\vec{a}_{avg.}$)

ball in air
(only mg is acting)

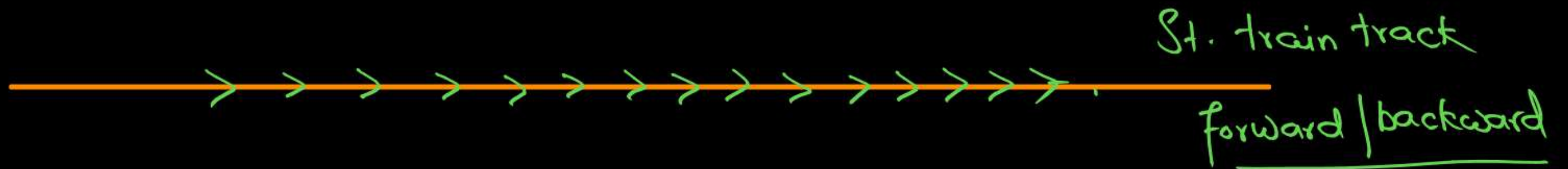


Velocity is the motion

acc. (force) change velocity

1-D Motion

Motion on St. line



Sign of displacement, Velocity and acceleration

(because they are vectors)

- Sign of displacement, velocity and acceleration only gives direction
 $\begin{matrix} \uparrow \\ -\downarrow \end{matrix} \rightarrow \begin{matrix} + \\ - \end{matrix} \} \rightarrow \underline{x\text{-axis}}$
- 1st of all we must decide which direction is +ive
 $\xrightarrow{\text{+ive}}$
 $\uparrow \text{+ive}$
- If quantity is in same direction which we assume +ive then that quantity have +ive sign
- If quantity is in opposite direction to which we assume +ive then that quantity have -ive sign

+ive (a, v, s $\rightarrow$ vector in 1D)

Sign of displacement, Velocity and acceleration



s, v, a +ive
→

↑
 s, v, a +ive

s, v, a -ive
←

↓
 s, v, a -ive

Question



A particle move from point A to point B. find displacement (S) of particle with sign and distance travelled (l)

$S = +30\text{m}, l = 30$
Vector

$S = -30, l = 30$

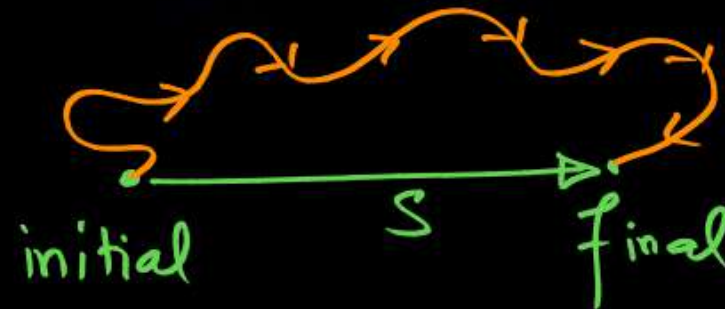


Question



A particle move from point A to point B. find displacement (S) of particle with sign and distance travelled (l)

displacement



$$S = +20, \quad l = 30 + 10 = 40$$



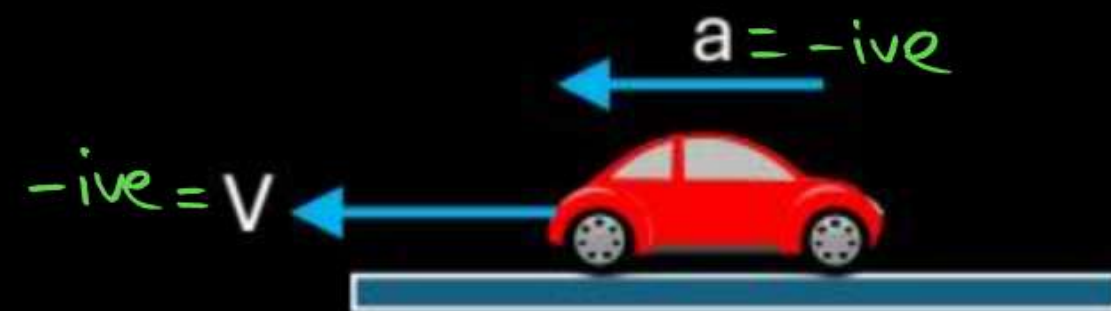
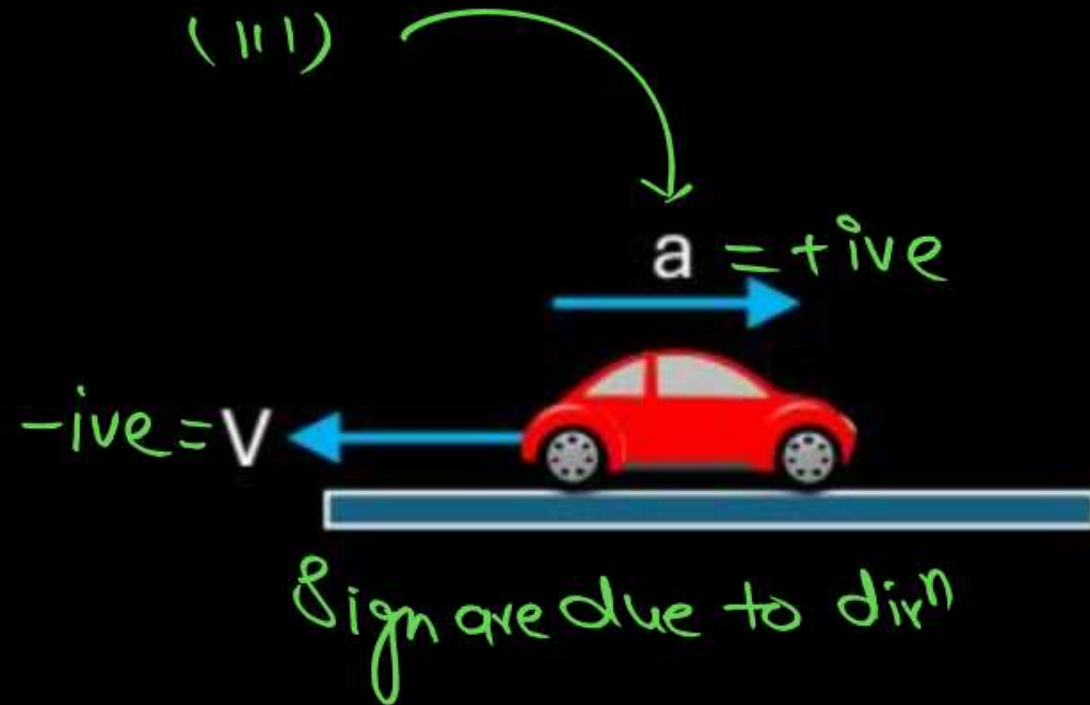
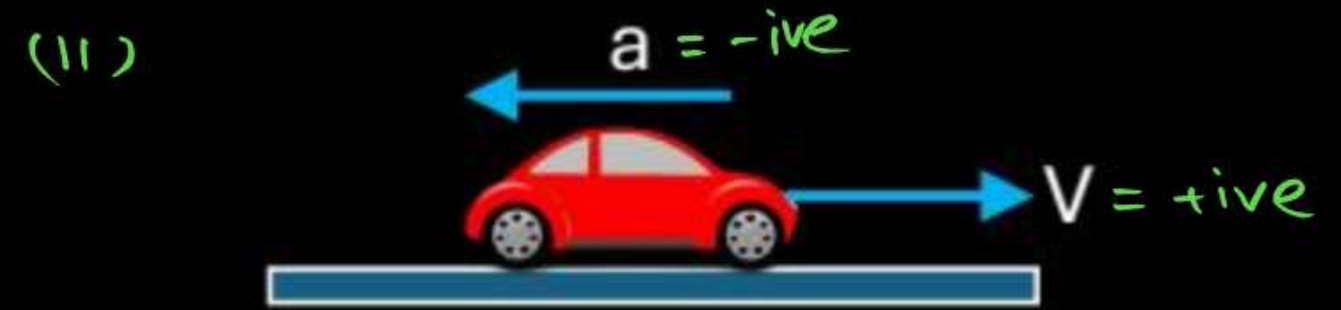
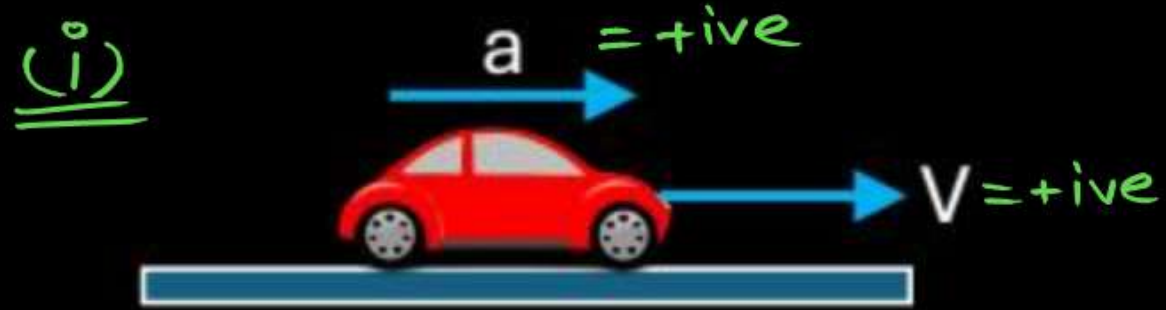
$$S = -10, \quad l = 30 + 40 = 70$$

Question



→ +ive

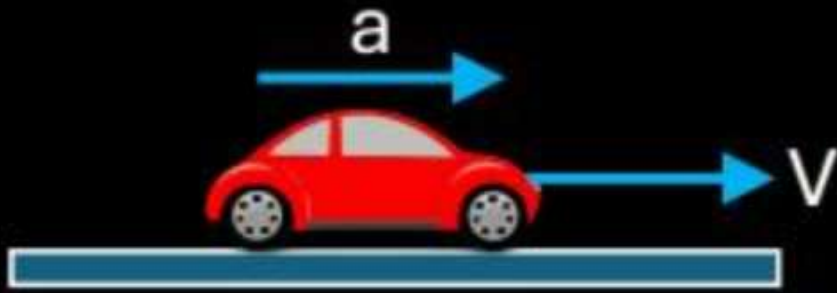
Velocity and acceleration of car are as shown. Find sign of velocity and acceleration



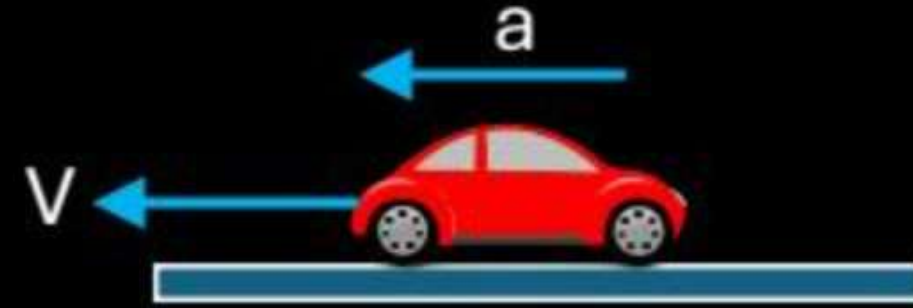
Speed up (Accelerated) or Speed down (Retarded)



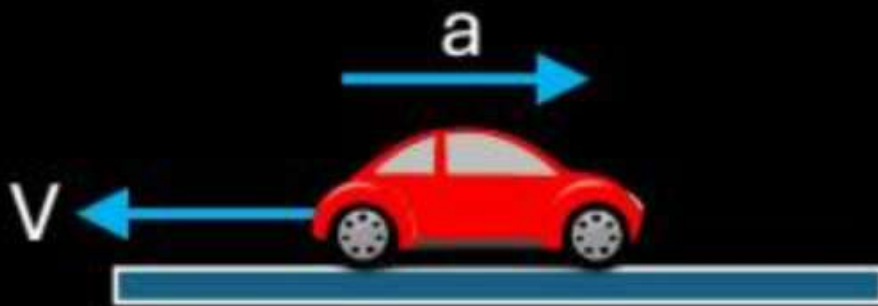
If velocity and acceleration are in same direction then particle is **speed up** (accelerated)



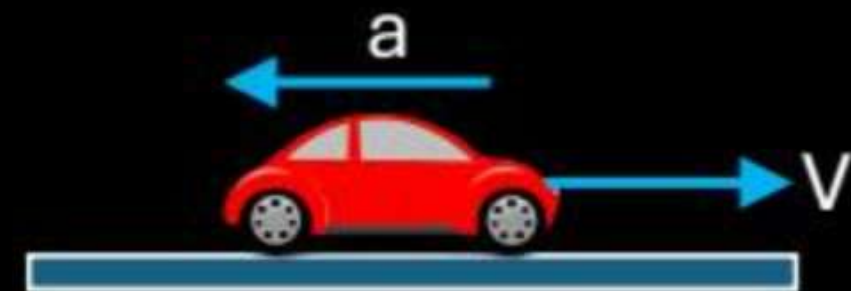
$$a = \frac{dv}{dt}$$

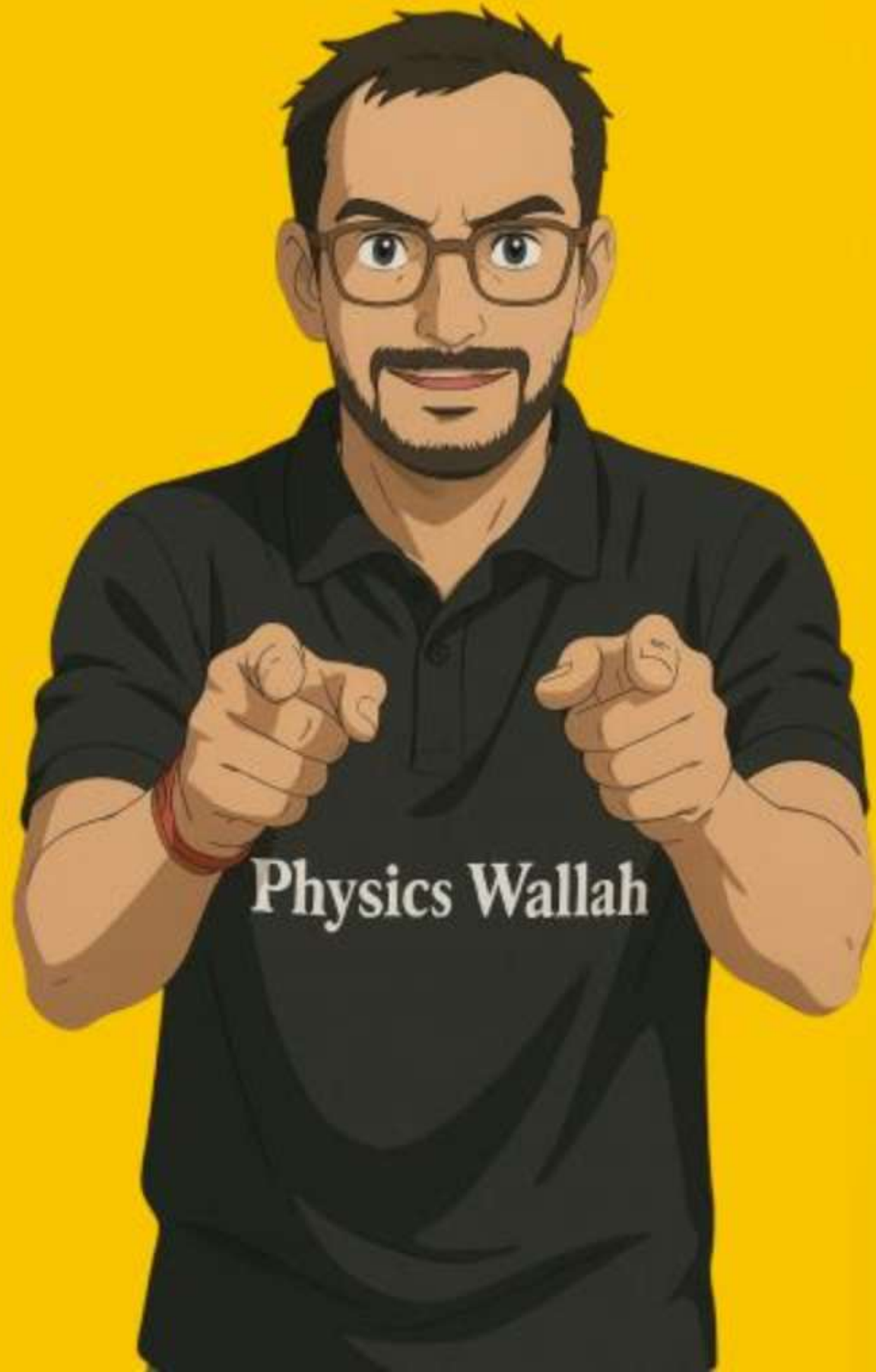


If velocity and acceleration are in opposite direction then particle is **speed down** (Retarded)



$$a = -\frac{dv}{dt}$$





THANK YOU
BAWWAL
BACCCHA
PARTY

